

Mechanisms determining species dominance in a phytoplankton bloom induced by the iron fertilization experiment EisenEx in the Southern Ocean

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Abstract

The dynamics of phytoplankton species populations recorded during the 3-week, iron-fertilization experiment EisenEx carried out in spring in the Antarctic Polar Frontal Zone are presented and discussed as the difference between growth and mortality rates. Only two cosmopolitan diatom species, the centric *Chaetoceros debilis* and the pennate *Pseudo-nitzschia lineola*, increased population density exponentially throughout the experiment to 150- and 90-fold of initial values, respectively. Because *C. debilis* initial abundance was tenfold lower than that of *P. lineola*, the two contributed 1% and 21% to bloom biomass, respectively at the end of the experiment, high-lighting the role of seeding in bloom formation. The other significant species increased population size at a linear rate throughout the experiment or for a short spurt phase to 3- to 18-fold of initial values. Conservative estimates of mortality rates within diatom species populations were obtained by comparing net accumulation rates of full cells with those of empty and broken frustules. The ratios were consistent over time for the various species but varied widely between them. The species-specific variation can be explained by differences in both growth and mortality rates, the latter partly due to either selective grazing or avoidance by the large protozo- and metazooplankton populations present. Selective predation by the abundant copepod populations on protistan grazers of diatoms (ciliates and heterotrophic dinoflagellates) apparently aided diatom biomass build-up. The response patterns of populations of the phytoplankton species present fall into six categories comprising disparate species, indicating that phylogeny is a poor predictor of ecology. The group that did not respond to fertilization was the most diverse and included both endemic and cosmopolitan as well as background and bloom-forming species. This lack of response to the advent of favorable growth conditions indicates that proximate factors during EisenEx triggered growth only in some species but had little effect on others. We attribute the differences in behavior to ultimate factors such as seasonal effects on life cycles and other internal constraints on growth rates. The implications for our understanding of the evolutionary ecology of phytoplankton and its impact on global biogeochemical cycles are pointed out.

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1. Introduction

Phytoplankton blooms play a crucial role in pelagic food webs, transfer of organic matter to the sea floor and ocean biogeochemistry. The fate of bloom biomass is strongly influenced by its species composition (Smetacek et al., 2004); however, the mechanisms regulating species dominance in blooms are poorly understood. Phytoplankton biomass accumulation in the surface layer is a function of integrated growth rates of species' populations in the surface mixed layer on the one hand and of their loss rates via sinking and mortality due to cell death, pathogens and predators (grazers) on the other. Whereas the bottom-up factors (light and nutrient supply) determining the growth environment of phytoplankton blooms are fairly well understood (Falkowski et al., 1998), comparable knowledge on the impact of the mortality environment (top-down factors) on species' populations is scant (Smetacek et al., 2004). In contrast to the growth environment, it is impossible to adequately simulate the mortality environment under laboratory conditions, but in situ iron fertilization experiments now provide the means to unravel the relative roles of the various processes determining growth and mortality of species' populations under natural settings.

All five iron fertilization experiments carried out in high-nutrient, low-chlorophyll (HNLC) waters of the Antarctic Circumpolar Current (ACC) induced phytoplankton blooms dominated by common ACC species (Boyd et al., 2000; Gervais et al., 2002; Coale et al., 2004; Hoffmann et al., 2006). The species composition of these artificially iron-fertilized blooms is similar to oceanic blooms fertilized by natural sources of iron along oceanic fronts (Smetacek et al., 2002), and from melting icebergs discharging their accumulated dust load in surface water (Smetacek et al., 1997). Dust precipitation is an additional, local and ephemeral source of iron likely to induce increases in phytoplankton biomass as observed in the HNLC North Pacific (Bishop et al., 2002). Since iron 'injections' to ACC surface waters are widespread phenomena, it follows that the species-specific responses reported here from the EisenEx experiment are representative of natural assemblages and can provide insights into pelagic ecosystem functioning not accessible by other methods.

We recorded net accumulation rates of dominant species populations of microphytoplankton during a bloom that developed in an 80 m deep mixed layer

to which dissolved iron was added in weekly intervals. Species-specific patterns in the growth and mortality environment of dominant diatom species were estimated from the concomitant net accumulation rates of full cells and empty, intact as well as broken frustules. The experiment was carried out for 3 weeks within a stable eddy in the Antarctic Polar Frontal Zone in the period when the phytoplankton in the region was undergoing its annual vernal increase (November). Much the same species increased their numbers inside and outside the fertilized patch, albeit to different levels.

Our study was part of a comprehensive interdisciplinary investigation of the evolution of the patch (Cisewski et al., 2005), changes in its carbon budget (Bakker et al., 2005), dynamics of primary production (Gervais et al., 2002) and populations of pico- and nanoplankton by flow cytometry (Veldhuis and Timmermans, 2001), bacterioplankton composition and abundance (Arrieta et al., 2004), and a wide range of grazers from ciliates and heterotrophic dinoflagellates to large copepods as well as copepod larvae (Henjes et al., 2007). Grazing rates of protists and copepods on diatoms were estimated with incubation experiments (Henjes et al., 2007; Schultes et al., 2006). The conclusions we draw are based on this comprehensive assessment of the processes that shaped the growth and mortality environments of the bloom-forming diatoms. Our results provide unique information on mechanisms underlying species dominance in phytoplankton blooms.

2. Material and methods

The in situ mesoscale iron fertilization experiment EisenEx was conducted in the Atlantic Sector of the Southern Ocean (21°E, 47°S) in austral spring (8–29 November 2000) during cruise ANT XVIII/2 of R.V. *Polarstern*. A cyclonic eddy (approximately 120 km wide) shed from the Antarctic Polar Front (APF) was chosen as the experimental site and its approximate center marked with a drifting buoy (Cisewski et al., 2005). An area of about 50 km² around the buoy was fertilized with 4 tons of iron sulfate in acidified solution (FeSO₄) on each of three occasions at intervals of 8 days (Bathmann and Smetacek, 2001). Sulfurhexafluoride (SF₆) was added as an inert tracer to the first iron infusion. Mapping of surface properties (especially of the inert tracer SF₆) was performed regularly in order to keep track of the expanding, fertilized "patch".

The area of the SF₆-marked fertilized patch increased by a factor of 20 while rotating within the eddy and covered much of the area of its core by the end of the experiment (Bakker et al., 2005). Hence, the growing bloom in the fertilized patch was diluted with surrounding water with its iron-limited, impoverished plankton community. Horizontal dispersion was initially high at 43% d⁻¹ but decreased after day 6 of the experiment and reached 5% d⁻¹ by the end of the experiment (Bakker et al., 2005). The average dilution rate of ~0.2 d⁻¹ we derive for the region of highest biomass we sampled is therefore conservative. Dilution effects due to diffusion affect all species equally as long as no significant changes occur outside of the fertilized patch. The dilution rate increases with distance from the center of the fertilized patch, but due to distortion in the course of rotation within the eddy, the area least influenced by dilution with outside water will not necessarily be located in the center of the patch (Turner et al., 2004) but will have the highest chlorophyll concentrations and F_v/F_m ratios. Therefore stations representative of the developing bloom (the 'in stations') were placed in the region of highest chlorophyll concentrations and F_v/F_m ratios within the SF₆-marked patch, which we located by underway mapping prior to sampling. This 'hot spot' within the patch also received subsequent iron additions of the same areal extent as the initial one. The closed heat and fresh water budget indicates that loss of fertilized water via lateral advection out of the patch was minor over the course of the experiment (Strass, V. H., pers. commun.) and presumably had little impact on the plankton assemblage.

Out-patch stations were located in adjacent waters with background SF₆ concentrations. Out-patch stations in the first half of the experiment were located within the eddy core, hence directly comparable to in-patch stations. However with increasing size of the patch due to lateral mixing, water masses that differed in several of the chemical and biological parameters were sampled by out stations, therefore, concentration differences between outside stations do not necessarily reflect temporal trends in the same water mass as is definitely the case for the in-patch stations. Nevertheless, they provide a useful baseline to assess trends within the patch.

For quantitative assessment of the phytoplankton assemblage 200 ml water samples were obtained from Niskin bottles attached to a conductivity

temperature depth (CTD) rosette from seven discrete depths between 10 and 150 m at 11 in and 5 out patch stations. Water samples were preserved with hexamine buffered formalin solution at a final concentration of 2% and stored at 4 °C in the dark for subsequent counting back in the home laboratory. Cells were identified and enumerated using inverted light and epifluorescence microscopy (Axiovert 25 and Axiovert 135, Zeiss, Oberkochen, Germany) according to the method of Utermöhl (1958). Water samples were settled in 50 ml sedimentation chambers (Hydrobios, Kiel, Germany) for 48 h. Organisms were counted at magnifications of 200–640 × according to the size of the organisms examined. Very abundant species were counted in transects, abundant species in a quarter or half a chamber and less abundant species in the whole chamber. Each sample was examined until at least 400 cells had been counted. In order to collect reliable numbers of *Phaeocystis antarctica* colonies the total content of a Niskin bottle (approximately 12 l) was concentrated to a volume of 50 ml by pouring the water gently through a 10 µm mesh net. Concentrated samples were fixed with hexamine buffered formalin solution at a final concentration of 0.5%. Three ml of each sample were counted at a magnification of 200 × in the whole chamber. Cells per colony were estimated from colony size and recognizable cells inside the colony.

Fine-meshed hand nets (20 and 55 µm) were used at each in and out patch station to collect plankton samples for preparations of diatom frustules for species identification. Of the 52 taxa that could be differentiated under light microscopy 49 were identified to species and three to genus level. Centric diatoms of disc-like shape that could not be assigned to a genus with certainty under light microscopy were grouped according to size. A small *Nitzschia*-like pennate diatom that could not be identified with certainty was categorized as an unidentified small pennate diatom. Although coccolithophores could not be distinguished to species level under light microscopy, scanning electron microscopy (SEM) investigations revealed only one species: *Emiliania huxleyi*. Photosynthetic dinoflagellates counted during this study included thecate dinoflagellates only. With the exception of the genus *Prorocentrum* and the species *Mesoporus perforatus* no further taxonomic identification was possible under light microscopy. Furthermore, athecate and thecate heterotrophic dinoflagellates as well as aloricate ciliates were counted in the same

samples but are dealt with by Henjes et al. (2007). Only intact empty and broken skeletons of the silicoflagellate *Dictyocha speculum* were encountered most likely because of disruption of the flagellate cell by the formalin fixation method. Intact empty skeletons were therefore assumed to have been alive prior to fixation which likely results in an over-estimation of the actual population size. In order to acquire information on species-specific mortality recognizable empty and broken diatom frustules were also counted in the same samples. Broken and empty diatom frustules were identified to species or genus level and otherwise grouped into size classes. Only broken diatom frustules of which >50% of the frustule was intact were counted. Broken frustules of delicate species like *Cylindrotheca closterium*, *Chaetoceros neglectus* and *C. debilis* were difficult to identify under light microscopy because their frustules tend to disintegrate into small debris that cannot be satisfactorily assigned to a single cell. Further intact empty frustules of these delicate species were rarely encountered in water samples so have not been considered here.

The cell size of the different species and groups was determined and their biovolume calculated from equivalent geometrical shapes (Edler, 1979). Cell volume was converted to cellular carbon content through recommended carbon conversion equations using the following carbon to volume relationships (Menden-Deuer and Lessard, 2000): for small diatoms (<3000 μm^3), $C \text{ cell}^{-1} = 0.288 \times V^{0.811}$; for large diatoms (>3000 μm^3), $C \text{ cell}^{-1} = 0.117 \times V^{0.881}$; for prymnesiophytes, $C \text{ cell}^{-1} = 0.228 \times V^{0.899}$; for dinoflagellates, $C \text{ cell}^{-1} = 0.444 \times V^{0.864}$ and for chrysophytes, $C \text{ cell}^{-1} = 0.020 \times V^{1.218}$, with V representing total cell volume (μm^3) and C the estimated cellular carbon content (pg). The mucous carbon of *Phaeocystis* colonies was calculated using the relationship $\text{ng C mucus} = 213 \times (\text{colony volume in mm}^3)$ (Mathot et al., 2000). *Phaeocystis* colonies in the samples were spherical in shape and ranged from 20 to 300 μm in diameter hence an average colony diameter of 160 μm was assumed.

Abundance and biomass of all species described herein are trapezoidal depth-integrated values for the upper 0–80 m of the water column

$$N_t = \left(\left(\int_0^{80} N \, dz \right) / z \right), \quad (1)$$

where N_t is the average abundance (biomass) at time t , $\int_0^{80} N$ the depth integral and z the depth (80 m).

Abundances are not given perm^{-2} but for a better comparison as the mean cell density l^{-1} in the upper 80 m.

Net accumulation rates—the difference between growth and mortality rates—were derived for all important diatom species and groups for the period of spurt growth. Net population accumulation rates (k) were calculated from integrated abundances according to Frost (1972)

$$k = \ln(N_1/N_0)/(t_1 - t_0), \quad (2)$$

where N_0 is the initial abundance at t_0 and N_1 the abundance at time t_1 . The actual gross accumulation rates (net growth rates) are then given by the net accumulation rates corrected for dilution.

Diatom mortality (g) derived from empty, intact and broken frustules (EBF) was calculated as follows:

$$g = ((N_{\text{EBF } t_1} - N_{\text{EBF } t_0})k)/(N_1 - N_0), \quad (3)$$

where $N_{\text{EBF } t_0}$ is the initial abundance of EBF at t_0 , $N_{\text{EBF } t_1}$ the abundance of EBF at t_1 , k the net accumulation rate of full cells (FC).

All data are archived in the information system PANGAEA at doi:10.1594/PANGAEA.552191.

3. Results

3.1. Environmental setting

The fertilized patch increased in area from initially 50 km^2 to nearly 1000 km^2 on day 21 by horizontal mixing (Turner et al., 2004). The seasonal mixed layer varied between 20 and 80 m depending on wind speed during the course of the experiment. Three storms in the second half of the experiment with gale force winds (up to 25 m s^{-1}) maintained the deep, seasonal mixed layer at 80 m (Cisewski et al., 2005). The euphotic depth steadily decreased inside the patch from initially >60 m to 41 m by the end of the experiment because of the evolving bloom (Gervais et al., 2002). Prior to fertilization ambient dissolved iron (< 0.2 μM) concentrations <0.1 nM (Nishioka et al., 2005) limited phytoplankton growth rates. After each iron infusion detectable Fe(II) concentrations lasted for up to 8 days (Croot et al., 2005). Concentrations of all major nutrients decreased during the experiment with nitrate by $\sim 2 \mu\text{M}$, phosphate by $\sim 0.2 \mu\text{M}$ and silicic acid by $\sim 4 \mu\text{M}$ but were still high by day 21 inside the patch (>21 μM nitrate, >1.5 μM

phosphate, $>10\ \mu\text{M}$ silicic acid; average over 80 m depth) (Gervais et al., 2002).

The areal daily primary production increased inside the patch and reached its maximum of $790\ \text{mg C m}^{-2}\text{d}^{-1}$ on day 16, and decreased thereafter (Gervais et al., 2002). The potential photosynthetic efficiency (F_v/F_m) measured with a fast repetition rate fluorometer (FRRF) increased within the patch from initially 0.3 to values as high as 0.56 with the strongest relative increase in the large cell fraction ($>20\ \mu\text{m}$). Chlorophyll *a* (Chl-*a*) concentrations increased five-fold inside the patch within 3 weeks (Gervais et al., 2002). Transmission values co-varied with Chl-*a* fluorescence both indicating the increase in phytoplankton concentrations inside the patch. In contrast to the iron-mediated increase in diatom cell numbers the abundance of the $<5\ \mu\text{m}$ phytoplankton fraction remained more or less constant over the course of the experiment (Veldhuis and Timmermans, 2001). Corresponding values outside the patch also remained more or less constant.

3.2. Temporal development of the phytoplankton assemblage

Inside the fertilized patch chlorophyll concentrations (Gervais et al., 2002) and phytoplankton carbon (PPC) derived from cell counts increased steadily through the 21 days of the experiment from initially 0.5 and 11 to 2.5 and $54\ \text{mg m}^{-3}$, respectively. The 80 m depth integrated chlorophyll and PPC stocks at the end of the experiment were $206\ \text{mg Chl m}^{-2}$ and $4.3\ \text{g C m}^{-2}$ (Fig. 1a) which are well within the range of typical phytoplankton blooms. Diatom biomass in the patch increased five-fold from initially $676\ \text{mg C m}^{-2}$ to $3475\ \text{mg C m}^{-2}$ (Fig. 1b) and accounted for 82% of the PPC at the end of the experiment. The dominant diatom species at the start of the experiment (discoid diatoms, *Dactyliosolen antarcticus*, *Fragilariopsis kerguelensis*, *Haslea trompii*, *Corethron pennatum* and *Guinardia delicatula*) taken together contributed roughly 50% to PPC, but by the end, *Pseudo-nitzschia lineola* achieved the dominant status, contributing 21% to PPC. The total non-diatom phytoplankton biomass increased from initially $258\ \text{mg C m}^{-2}$ to $944\ \text{mg C m}^{-2}$ on day 11 and declined thereafter to $777\ \text{mg C m}^{-2}$ on day 21 (Fig. 1c). *Prorocentrum* spp. contributed most to biomass of the non-diatom phytoplankton.

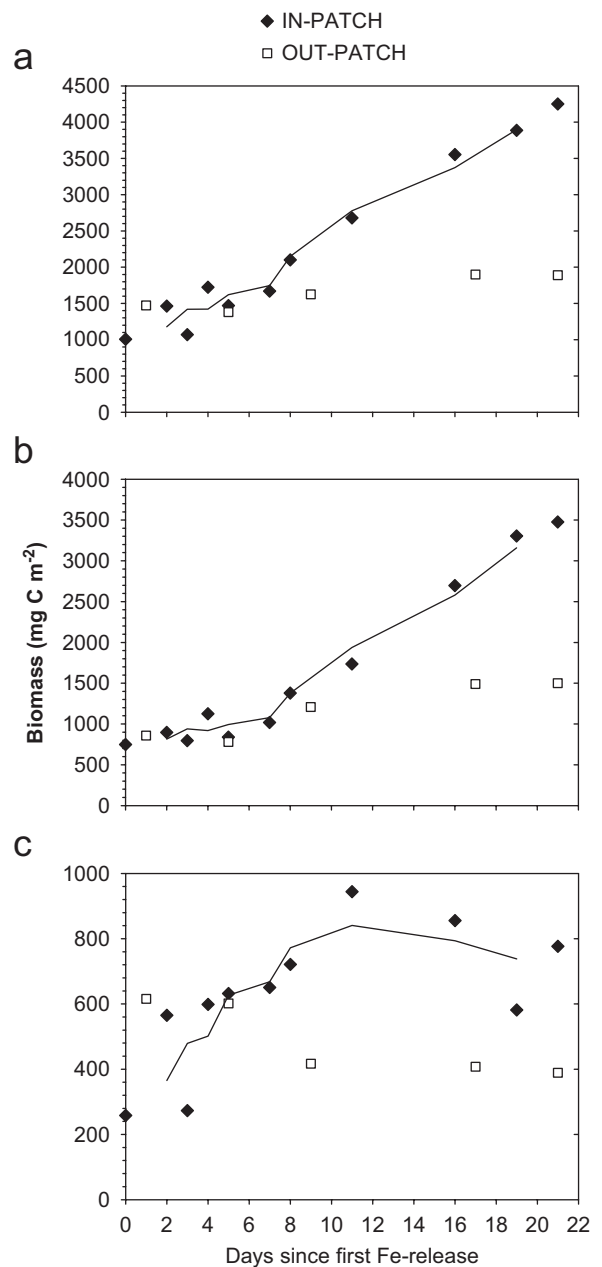


Fig. 1. Temporal development of 80 m-depth integrated biomass of (a) total phytoplankton, (b) diatoms and (c) non-diatom phytoplankton. Symbols indicate samples taken inside (solid diamonds) and outside (open squares) the fertilized patch. The thin line represents the running average over three temporally adjacent in-patch stations.

Outside the patch chlorophyll stocks remained more or less constant and PPC increased from $1474\ \text{mg C m}^{-2}$ on day 1 to $1891\ \text{mg C m}^{-2}$ on day 21. Total diatom biomass roughly doubled from $858\ \text{mg C m}^{-2}$ on day 1 to $1603\ \text{mg C m}^{-2}$ on day 21

whereas total non-diatom phytoplankton biomass decreased from 616 mg C m^{-2} to 371 mg C m^{-2} after three weeks. The diatom biomass increase was mainly due to large cylindrical diatoms. *Prorocentrum* spp. was again the dominant non-diatom.

From day 16 onwards diatom biomass was significantly higher inside than outside the patch (unpaired *t*-test; $p < 0.05$). The increase occurred within the seasonal mixed layer and concentrations of full cells (FC) at 100 and 150 m decreased towards the end of the experiment to $< 10\%$ of values recorded in 80 m at the base of the pycnocline (Fig. 2a). We found no trend in FC concentrations inside the patch below 80 m. Empty but intact diatom frustules (EF) and broken diatom frustules (BF) on the other hand showed a continuous vertical distribution down to 150 m depth (Fig. 2b and c). A complete list of species with category affiliation, accumulation rate for the period of net increase and their contribution to PPC inside and outside the patch is given in Table 1.

Although many of the diatom species responded to iron fertilization by increasing population size, the range of responses differed widely amongst species. Six major categories were distinguished based on abundance patterns and net accumulation

rates of characteristic species over the course of the experiment. The species making up these categories apply only to the EisenEx bloom.

- Species with sustained exponential increase.
- Species that increased continuously albeit linearly throughout the experiment.
- Species that increased linearly but for only a short spurt phase.
- Species with an initial linear increase and a decline thereafter.
- Species that exhibited no consistent trend.
- Species with a continuous decline in abundance.

The first category was best represented by two species: the pennate *Pseudo-nitzschia lineola* and the centric *Chaetoceros debilis*. Both showed an immediate response to iron fertilization (insets in Figs. 3a and b) with an exponential increase in cell numbers and exhibited the highest observed net accumulation rates of 0.20 and 0.18 d^{-1} , respectively. *P. lineola* numbers increased 90-fold from initially 2.6×10^3 to $234 \times 10^3 \text{ cells l}^{-1}$ within 3 weeks accounting for 53% of total diatom abundance (Fig. 3a) and 25% of total diatom biomass at the end of the experiment. Outside the fertilized

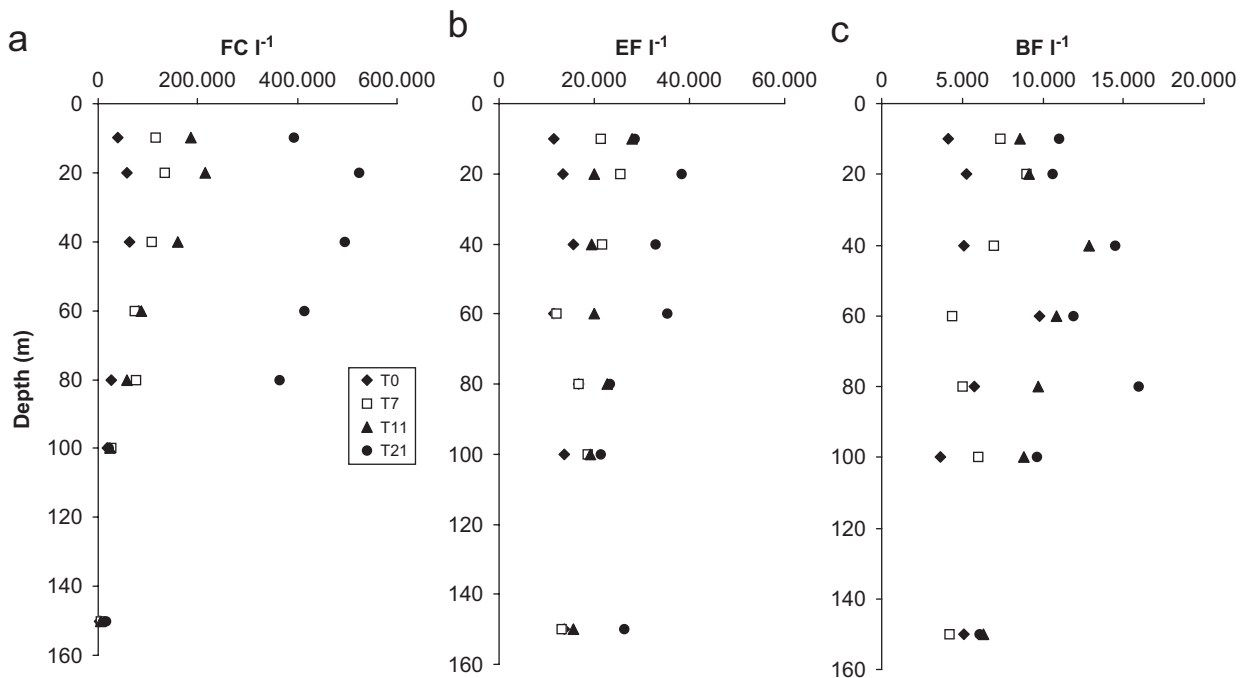


Fig. 2. Depth profiles of (a) full cells (FC), (b) intact empty (EF) and (c) broken diatom frustules (BF) at the start of the experiment (T0), after 7 (T7), 11 (T11) and 21 (T21) days inside the patch over the upper 150 m of the water column.

Table 1

Category affiliation and net accumulation rates for all species and groups identified during EisenEx. Net accumulation rates for categories 3 and 4 are only given for the time period of increase. Furthermore, the 80 m depth integrated PPC in mg Cm⁻² and relative contribution of each species and group (%) to the total PPC at the start (T0), after 11 (T11) and 21 (T21) days inside the fertilized patch as well as the mean of all outside stations is given

Taxon	Category	Accumulation rate	T0		T11		T21		Mean out-patch	
			PPC	(%)	PPC	(%)	PPC	(%)	PPC	(%)
Total phytoplankton biomass			1006.1		2679.2		4252.0		1653.5	
<i>Pseudo-nitzschia lineola</i> (Cleve) Hasle	1	0.20	9.0	0.9	93.7	3.5	883.1	20.8	38.6	2.3
<i>Corethron pennatum</i> (Castracane) Crawford	2	0.08	42.0	4.2	240.8	9.0	445.9	10.5	121.2	7.3
<i>Prorocentrum</i> spp.	4	0.12 (T0–T11)	35.9	3.6	558.5	20.8	395.4	9.3	197.1	11.9
Discoid diatoms < 30 µm	5	—	152.3	15.1	169.5	6.3	270.2	6.4	127.1	7.7
<i>Haslea trossii</i> (Cleve) Simonsen	2	0.08	46.0	4.6	126.0	4.7	226.5	5.3	45.4	2.7
<i>Dactyliosolen antarcticus</i> Castracane	3	0.11 (T11–T21)	117.0	11.6	126.6	4.7	209.5	4.9	149.6	9.0
<i>Proboscia alata</i> (Brightwell) Sundström	3	0.11 (T8–T19)	0.4	0.0	134.5	5.0	198.8	4.7	74.8	4.5
<i>Dictyocha speculum</i> Ehrenberg	2	0.03	50.4	5.0	146.0	5.4	162.9	3.8	132.1	8.0
Discoid diatoms > 30 µm	5	—	59.1	5.9	165.2	6.2	139.6	3.3	53.4	3.2
<i>Fragilariopsis kerguelensis</i> (O'Meara) Hustedt	3	0.15 (T7–T16)	46.6	4.6	91.5	3.4	137.3	3.2	47.7	2.9
Thecate dinoflagellates	2	0.04	59.0	5.9	113.2	4.2	125.5	3.0	51.3	3.1
<i>Pseudo-nitzschia heimii</i> Manguin	1	0.14	11.6	1.2	37.0	1.4	123.2	2.9	33.4	2.0
<i>Rhizosolenia chunii</i> Karsten	2	0.09	22.6	2.2	34.7	1.3	116.8	2.7	86.1	5.2
<i>Thalassionema nitzschioides</i> (Grunow) Grunow ex Hustedt	3	0.09 (T7–T21)	18.4	1.8	41.4	1.5	82.7	1.9	15.4	0.9
<i>Pseudo-nitzschia turgidula</i> (Hustedt) Hasle	1	0.13	7.6	0.8	26.0	1.0	82.2	1.9	12.7	0.8
<i>Guinardia cylindrus</i> (Cleve) Hasle	2	0.03	23.7	2.4	48.4	1.8	74.0	1.7	89.2	5.4
<i>Corethron inerme</i> Karsten	5	—	3.1	0.3	11.6	0.4	73.2	1.7	21.1	1.3
<i>Thalassiosira tumida</i> (Janisch) Hasle	Rare	—	7.3	0.7	59.3	2.2	61.0	1.4	10.5	0.6
<i>Emiliania huxleyi</i> (Lohmann) Hay & Mohle	6	−0.02	106.0	10.5	96.7	3.6	59.2	1.4	92.5	5.6
<i>Chaetoceros debilis</i> Cleve	1	0.18	0.3	0.0	10.4	0.4	45.4	1.1	1.2	0.1
<i>Pseudo-nitzschia turgiduloides</i> Hasle	1	0.14	6.6	0.7	13.1	0.5	42.3	1.0	5.3	0.3
<i>Asteromphalus hyalinus</i> Karsten	5	—	23.7	2.4	26.6	1.0	36.0	0.8	14.0	0.8
<i>Leptocylindrus mediterraneus</i> (Peragallo) Hasle	2	0.07	10.2	1.0	30.8	1.2	34.1	0.8	33.5	2.0
<i>Phaeocystis antarctica</i> (solitary) Karsten	4	0.07 (T0–T16)	5.6	0.6	28.4	1.1	32.8	0.8	12.4	0.8
<i>Membraneis imposter</i> Paddock	Rare	—	4.7	0.5	24.3	0.9	27.1	0.6	9.5	0.6
<i>Navicula</i> sp.	3	0.06 (T7–T21)	12.3	1.2	16.7	0.6	22.9	0.5	12.0	0.7
<i>Cylindrotheca closterium</i> (Ehrenberg) Lewin & Reimann	4	0.08 (T0–T16)	7.1	0.7	21.2	0.8	21.1	0.5	12.0	0.7
<i>Asteromphalus hookeri</i> Ehrenberg	5	—	3.4	0.3	29.5	1.1	13.3	0.3	11.8	0.7
<i>Thalassiosira lentiginosa</i> (Janisch) Fryxell	5	—	19.1	1.9	26.4	1.0	12.0	0.3	24.2	1.5
<i>Chaetoceros convolutus</i> Castracane	1	0.11	1.0	0.1	4.8	0.2	11.8	0.3	3.0	0.2
<i>Thalassiothrix antarctica</i> (Schimper) Karsten	2	0.05	1.7	0.2	9.4	0.3	11.3	0.3	20.9	1.3
<i>Rhizosolenia antennata</i> f. <i>semispina</i> Sundström	Rare	—	4.7	0.5	3.5	0.1	11.1	0.3	8.0	0.5
<i>Thalassionema nitzschioides</i> var. <i>lanceolatum</i>	5	—	3.4	0.3	5.5	0.2	9.2	0.2	4.9	0.3
<i>Azpeitia tabularis</i> (Grunow) Fryxell & Sims	4	0.07 (T2–T8)	13.7	1.4	12.7	0.5	8.9	0.2	10.4	0.6
<i>Thalassiosira</i> cf. <i>oliverana</i> (O' Meara) Mak	5	—	4.2	0.4	8.7	0.3	7.4	0.2	3.7	0.2
<i>Thalassiosira oestrupii</i> (Ostenfeld) Hasle	5	—	4.9	0.5	5.6	0.2	6.8	0.2	4.0	0.2
<i>Thalassiosira gracilis</i> (Karsten) Hustedt	4	0.06 (T0–T11)	7.5	0.7	14.5	0.5	6.8	0.2	8.5	0.5
<i>Guinardia delicatula</i> (Cleve) Hasle	5	—	30.6	3.0	46.0	1.7	5.7	0.1	20.5	1.2
<i>Chaetoceros dichæta</i> Ehrenberg	5	—	0.0	0.0	0.0	0.0	4.5	0.1	0.8	0.1
<i>Chaetoceros atlanticus</i> Cleve	5	—	1.7	0.2	2.1	0.1	3.6	0.1	3.0	0.2
<i>Chaetoceros neglectus</i> Karsten	5	—	0.8	0.1	1.9	0.1	1.8	0.0	1.1	0.1
<i>Chaetoceros aequatorialis</i> Cleve	5	—	0.4	0.0	0.7	0.0	1.2	0.0	1.6	0.1
<i>Pleurosigma</i> cf. <i>directum</i> Cleve & Grunow	Rare	—	11.5	1.1	5.7	0.2	1.2	0.0	3.5	0.2
<i>Chaetoceros peruvianus</i> Brightwell	4	0.14 (T0–T7)	1.2	0.1	1.2	0.0	1.2	0.0	1.5	0.1
<i>Fragilariopsis obliquecostata</i> (Van Heurck) Hasle	Rare	—	2.0	0.2	0.6	0.0	1.1	0.0	0.0	0.0
<i>Pseudo-nitzschia prolongatoides</i> (Hasle) Hasle	4	0.05 (T0–T16)	2.1	0.2	4.1	0.2	1.1	0.0	4.5	0.3
<i>Mesoporos perforatus</i> (Gran) Lillick	Rare	—	1.4	0.1	1.6	0.1	1.1	0.0	0.9	0.1
<i>Eucampia antarctica</i> (Castracane) Mangin	Rare	—	0.3	0.0	0.0	0.0	0.8	0.0	1.6	0.1

Table 1 (continued)

Taxon	Category	Accumulation rate	T0		T11		T21		Mean out-patch	
			PPC	(%)	PPC	(%)	PPC	(%)	PPC	(%)
Small pennate diatom	4	0.03 (T0–T11)	0.7	0.1	1.2	0.0	0.5	0.0	0.9	0.1
<i>Chaetoceros</i> spp.	6	–0.07	1.0	0.1	0.5	0.0	0.2	0.0	0.4	0.0
<i>Nitzschia bicaipitata</i> Cleve	Rare	—	0.1	0.0	0.1	0.0	0.2	0.0	0.0	0.0
<i>Fragilariopsis rhombica</i> (O'Meara) Husted	6	–0.11	0.5	0.1	0.8	0.0	0.0	0.0	0.7	0.0
<i>Rhizosolenia curvata</i> Zacharias	Rare	—	0.1	0.0	0.1	0.0	0.2	0.0	0.0	0.0
<i>Chaetoceros bulbosus</i> (Ehrenberg) Heiden	Rare	—	0.1	0.0	0.1	0.0	0.2	0.0	0.0	0.0
<i>Actinocyclus curvatulus</i> Janisch	Rare	—	0.1	0.0	0.1	0.0	0.2	0.0	0.0	0.0

patch abundances of *P. lineola* also increased albeit slightly at net accumulation rates of 0.09 d^{-1} . *C. debilis* showed an 150-fold increase from initially 0.1×10^3 to $19 \times 10^3 \text{ cells l}^{-1}$ on day 21 inside the patch (Fig. 3b) but remained low outside it.

Corethron pennatum and *Haslea tromptii* were the major representatives of the second category with a continuous linear increase in cell numbers after iron addition with net accumulation rates of 0.08 d^{-1} . *C. pennatum* increased from initially 0.5×10^3 to $4.6 \times 10^3 \text{ cells l}^{-1}$ by day 21 inside the patch (Fig. 3c). Outside the patch cell numbers increased from $1.1 \times 10^3 \text{ cells l}^{-1}$ on day 1 to $1.7 \times 10^3 \text{ cells l}^{-1}$ on day 21, corresponding to a net accumulation rate of 0.04 d^{-1} . The pennate diatom *Haslea tromptii* showed a similar increase in cell numbers from initially 0.6×10^3 to $2.9 \times 10^3 \text{ cells l}^{-1}$ by day 21 (Fig. 3d). In out-patch waters abundances stayed more or less constant.

Fragilariopsis kerguelensis and *Thalassionema nitzschioides* belong to the third category. These heavily silicified species showed only very minor increases in abundance during the first week of the experiment. Thereafter both species showed a linear increase with net accumulation rates of 0.15 d^{-1} for *F. kerguelensis* and 0.09 d^{-1} for *T. nitzschioides*. Initial abundances of *F. kerguelensis* were relatively high with $5.7 \times 10^3 \text{ cells l}^{-1}$ and accounted for 10% of the total diatom assemblage at the start of the experiment. Cell numbers increased to $23 \times 10^3 \text{ cells l}^{-1}$ by day 16 inside the patch and stagnated thereafter (Fig. 3e). The relative contribution of *F. kerguelensis* to total diatom abundance had declined to 5% at the end of EisenEx. Outside the fertilized patch no consistent trend was observed and abundances varied between 4×10^3 and $9 \times 10^3 \text{ cells l}^{-1}$. At the start of the experiment *T. nitzschioides* was the second most abundant

diatom species with $11 \times 10^3 \text{ cells l}^{-1}$ accounting for 19% of the total diatom assemblage. Cell numbers of *T. nitzschioides* further increased to $48 \times 10^3 \text{ cells l}^{-1}$ by day 21 inside the patch although its relative importance decreased to 11% of the total diatom abundance (Fig. 3f). Outside the patch a reverse trend was observed with a decrease to $5 \times 10^3 \text{ cells l}^{-1}$ by day 21. Two large-celled diatom species also belonging to this category, *Dactylosolen antarcticus* and *Proboscia alata*, each contributed 5% to PPC inside the patch at the end of the experiment. Their abundances, however, were low.

Characteristic of category 4 were an unidentified, small pennate diatom, *Prorocentrum* spp. and *Phaeocystis antarctica*. The former 2 showed an initial linear increase until day 11 with net accumulation rates of 0.03 and 0.12 d^{-1} , respectively and a decline thereafter. Abundances of the unidentified, small pennate diatom, which were not significantly different inside and outside the patch, increased from initially 1.5×10^3 to $2.6 \times 10^3 \text{ cells l}^{-1}$ by day 11 and declined thereafter to values of $1.1 \times 10^3 \text{ cells l}^{-1}$ on day 21 (Fig. 3g). *Prorocentrum* spp. increased after iron addition from initially 0.88×10^3 to $13 \times 10^3 \text{ cells l}^{-1}$ on day 11. Thereafter cell numbers decreased to $5 \times 10^3 \text{ cells l}^{-1}$ on day 21 (Fig. 3h). Outside the patch depth integrated abundances varied between 2 and $6 \times 10^3 \text{ cells l}^{-1}$. The total abundance of *P. antarctica* inside the patch including both solitary and colonial cells, increased from initially 8.1×10^3 to $138 \times 10^3 \text{ cells l}^{-1}$ by day 16, thereafter cell numbers decreased to $49 \times 10^3 \text{ cells l}^{-1}$ by the end of the experiment (Fig. 3i). Outside the patch the depth integrated total abundances of *P. antarctica* showed no consistent trend and varied between 11×10^3 and $29 \times 10^3 \text{ cells l}^{-1}$. Solitary cells of *P. antarctica* dominated the total abundance by far. Cells in

colonies accounted on average for less than 2% of the total population both inside as well as outside the fertilized patch. Despite their relatively small contribution to the total abundance, cells in

colonies increased significantly inside the patch. Outside the patch both colonies and cells in colonies showed no increase over the duration of the experiment. Net accumulation rates for solitary

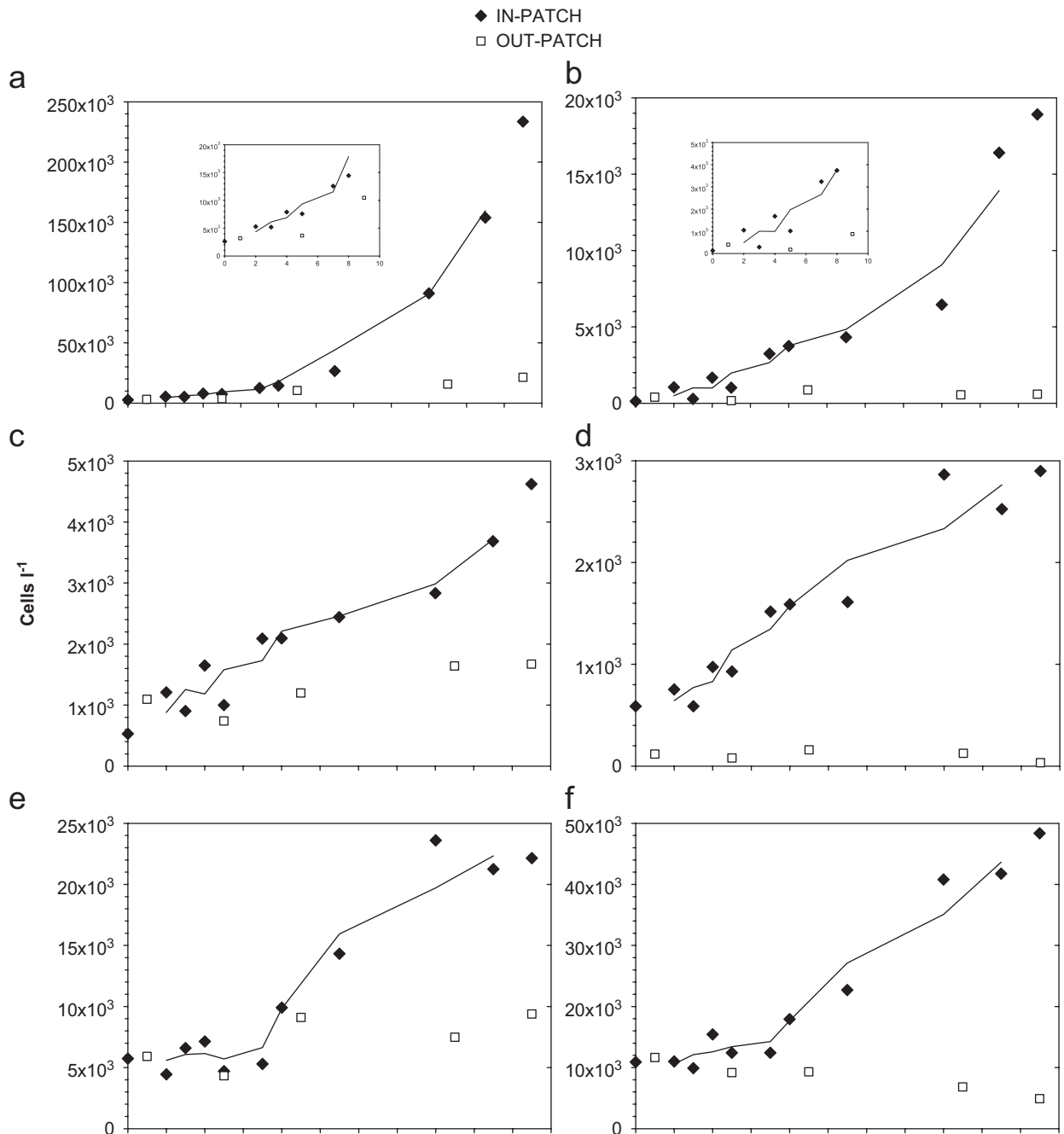


Fig. 3. Temporal development of 80 m-depth integrated abundance of (a) *Pseudo-nitzschia lineola*, (b) *Chaetoceros debilis*, (c) *Corethron pennatum*, (d) *Haslea trompii*, (e) *Fragilariopsis kerguelensis*, (f) *Thalassionema nitzschioides*, (g) unidentified small pennate diatom, (h) *Prorocentrum* spp., (i) *Phaeocystis antarctica* (j) *Asteromphalus hyalinus*, (k) *Thalassiosira lentiginosa* and (l) *Emiliania huxleyi*. Symbols and thin line as in Fig. 1.

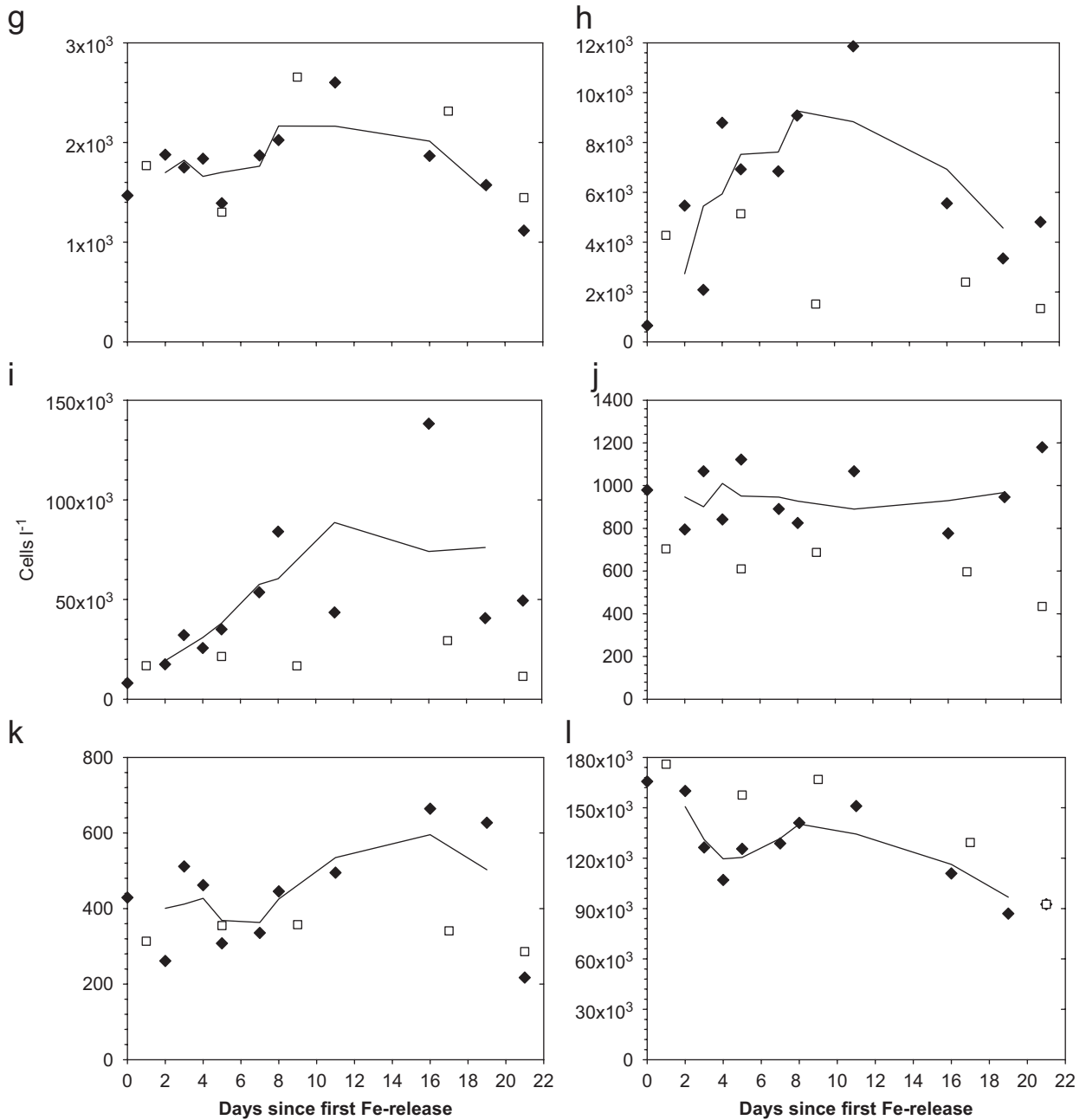


Fig. 3. (Continued)

cells, colonies and cells in colonies amounted to 0.07, 0.08 and 0.10 d⁻¹, respectively inside the patch.

Category 5 comprised a large variety of species from different genera that were a numerically minor component of the total assemblage and showed similar abundances inside and outside the patch with no consistent trend in response to iron addition. The two centric diatoms *Asteromphalus*

hyalinus and *Thalassiosira lentiginosa* belonged to this category. Both species were characterized by relatively stable populations over the course of the experiment inside as well as outside the patch. Abundances of *A. hyalinus* varied between 0.8–1.2 × 10³ cells l⁻¹ inside and 0.4–0.7 × 10³ cells l⁻¹ outside the patch (Fig. 3j). Population size of *T. lentiginosa* varied between 0.2–0.7 × 10³ cells l⁻¹

inside and $0.3\text{--}0.4 \times 10^3 \text{ cells l}^{-1}$ outside the patch (Fig. 3k).

The coccolithophorid *Emiliania huxleyi* was characteristic for category 6 with a consistent decline inside as well as outside the patch. Initial abundances were high with $166 \times 10^3 \text{ cells l}^{-1}$. Thereafter, cell numbers steadily decreased reaching $93 \times 10^3 \text{ cells l}^{-1}$ at the end of the experiment (Fig. 3l). Abundances inside and outside the patch were not significantly different.

Species that were not assigned to any of the above categories were rare, usually large in size and showed no consistent trend over the course of the experiment. Because of their low abundances (well below $0.5 \times 10^3 \text{ cells l}^{-1}$) these species were under-represented in water samples and have hence not been grouped under category 5.

3.3. Empty and broken diatom frustules

Stocks of both empty and broken diatom frustules (EBF) increased over the course of the experiment. EF more than doubled inside the patch from initially 14×10^3 to $32 \times 10^3 \text{ EF l}^{-1}$ by the end of the experiment whereas values outside the patch did not change (Fig. 4a). BF doubled inside the patch from initially 6.2×10^3 to $13 \times 10^3 \text{ BF l}^{-1}$ on day 21 and increased slightly outside the patch (Fig. 4b). Both EF and BF of *Pseudo-nitzschia lineola* showed a steep increase inside the patch from initially $0.3 \times 10^3 \text{ EF l}^{-1}$ and BF l^{-1} to $13 \times 10^3 \text{ EF l}^{-1}$ and $3.6 \times 10^3 \text{ BF l}^{-1}$ after 3 weeks, respectively (Fig. 5a). EF of *Corethron pennatum* were rare and varied between 0.01 and $0.1 \times 10^3 \text{ EF l}^{-1}$, whereas BF increased after iron addition from 0.2×10^3 to $0.8 \times 10^3 \text{ BF l}^{-1}$ (Fig. 5b). A doubling in EF of *Fragilariopsis kerguelensis* from initially 1.4×10^3 to $2.7 \times 10^3 \text{ EF l}^{-1}$ was observed. *F. kerguelensis* BF varied between 0.5×10^3 and $1.7 \times 10^3 \text{ BF l}^{-1}$ with a slight increase towards the end of the experiment (Fig. 5c). EF of *Thalassionema nitzschoides* increased from initially 2.7×10^3 to $6.5 \times 10^3 \text{ EF l}^{-1}$ by the end of the experiment. BF of *T. nitzschoides* showed a strong increase from $0.9 \times 10^3 \text{ BF l}^{-1}$ at the start to $3.1 \times 10^3 \text{ BF l}^{-1}$ by day 16 and a decline thereafter (Fig. 5d). EBF derived mortality rates ranged from $0.03\text{--}0.06 \text{ d}^{-1}$ in *P. lineola*, $0.01\text{--}0.05 \text{ d}^{-1}$ in *F. kerguelensis*, $0\text{--}0.006 \text{ d}^{-1}$ in *T. nitzschoides* and $0.01\text{--}0.04 \text{ d}^{-1}$ in *C. pennatum*. Despite the increase in absolute numbers of empty and broken diatom frustules the relative contribution of both fractions to the

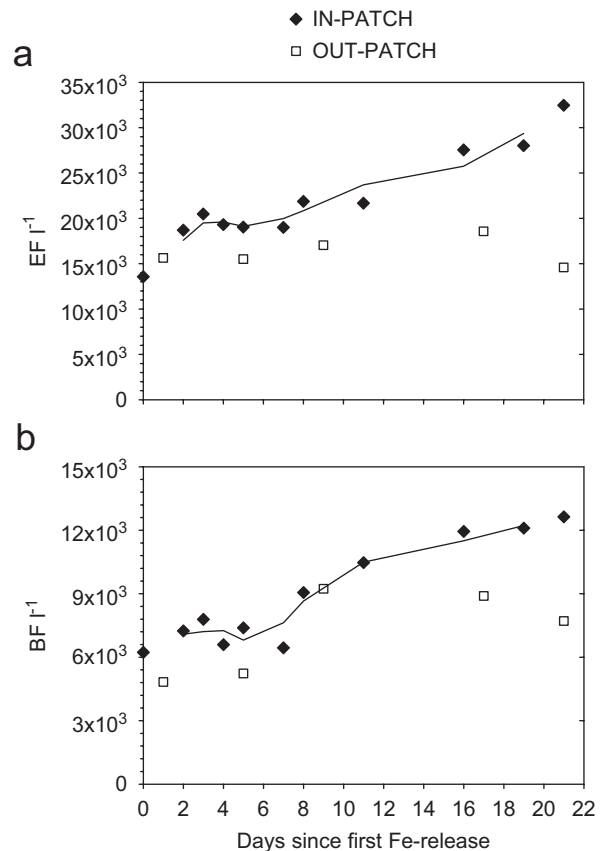


Fig. 4. Temporal development of 80m-depth integrated abundance of (a) intact empty (EF) and (b) broken diatom frustules (BF). Symbols and thin line as in Fig. 1.

total diatom assemblage decreased with time inside the patch because of the proportionally stronger increase of FC. Thus, the ratio of FC to empty and broken frustules (FC:EBF) increased from initially 2.9 to 9.9 inside and 4 outside the patch on day 21.

Species-specific differences in FC:EBF of significant species at the start and end of the experiment inside the fertilized patch ranged from 4.6 to 14.5 in *P. lineola* and from 1.8 to 5.2 in *C. pennatum*. The other significant species had ratios in between. Thus, the ratio increased about threefold in each of the significant species albeit from different levels. Interestingly, the ratios FC:EBF in two morphologically similar but background species of *Pseudo-nitzschia* differed significantly and consistently from *P. lineola* (Fig. 6): In *P. turgidula* FC:EBF increased from 2.8 at the start of the experiment to 12.2 on day 16 but decreased to 6.9 on day 21, whereas in *P. heimii* a high initial FC:EBF of 10.6 further increased to 25.8 at the end of the experiment.

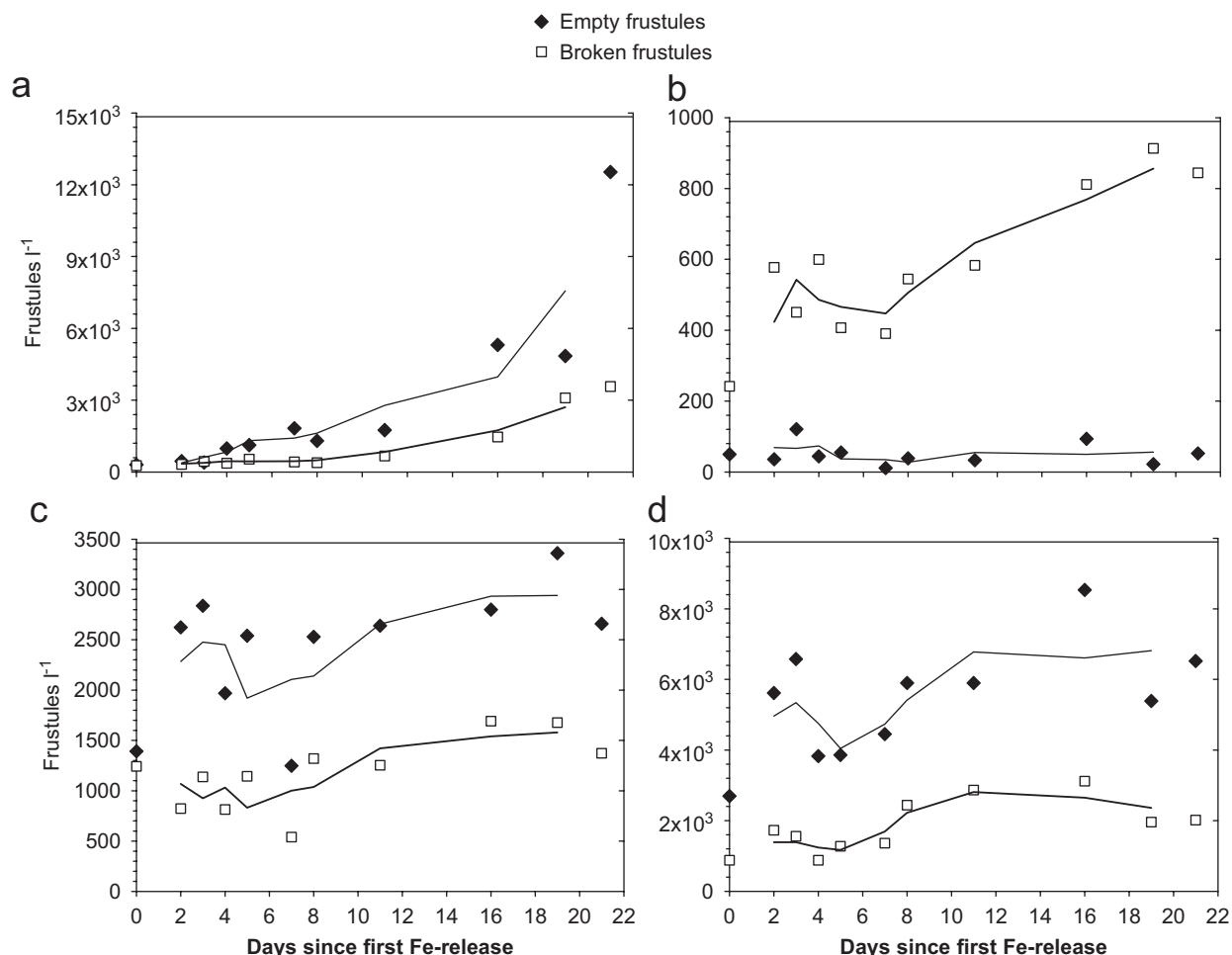


Fig. 5. Temporal development of 80m-depth integrated abundance of intact empty (EF) and broken diatom frustules (BF) inside the fertilized patch of (a) *Pseudo-nitzschia lineola*, (b) *Corethron pennatum*, (c) *Fragilariopsis kerguelensis* and (d) *Thalassionema nitzschioides*. Symbols indicate empty frustules (EF) (solid diamonds) and broken frustules (BF) (open squares). Thin line as in Fig. 1.

The ratio of empty to broken diatom frustules (EF:BF) for all diatoms combined showed little change over the course of the experiment and was on average 2.5 ± 0.3 . However, this ratio varied considerably and consistently between species presumably because of discrimination by grazer populations. The EF:BF ratio of *P. lineola* increased from initially 1.2 to 3.5 on day 21 whereas the reverse trend could be observed for *P. turgidula* from 5.0 on day 2 to 0.9 at the end of the experiment. *P. heimii* showed a variable EF:BF ratio over the course of the experiment with an average ratio of 1.0 ± 1.18 . EF:BF of *F. kerguelensis* and *T. nitzschioides* showed no consistent trend with time and exhibited average ratios of 2.1 ± 0.5 and 3.1 ± 0.7 , respectively. The EF:BF of *C. pennatum*

was an order of magnitude lower than all other species and decreased from initially 0.2 to 0.1 on day 21.

4. Discussion

The continuous, comprehensive observations of the developmental dynamics of an artificially enhanced spring bloom presented here provide unique insights into the ecology of open ocean phytoplankton. In the following, we first introduce the major diatom species that responded to iron fertilization by highlighting their differences. We next discuss the roles of bottom-up versus top-down factors in shaping composition of the growing bloom from which we draw some obvious and some

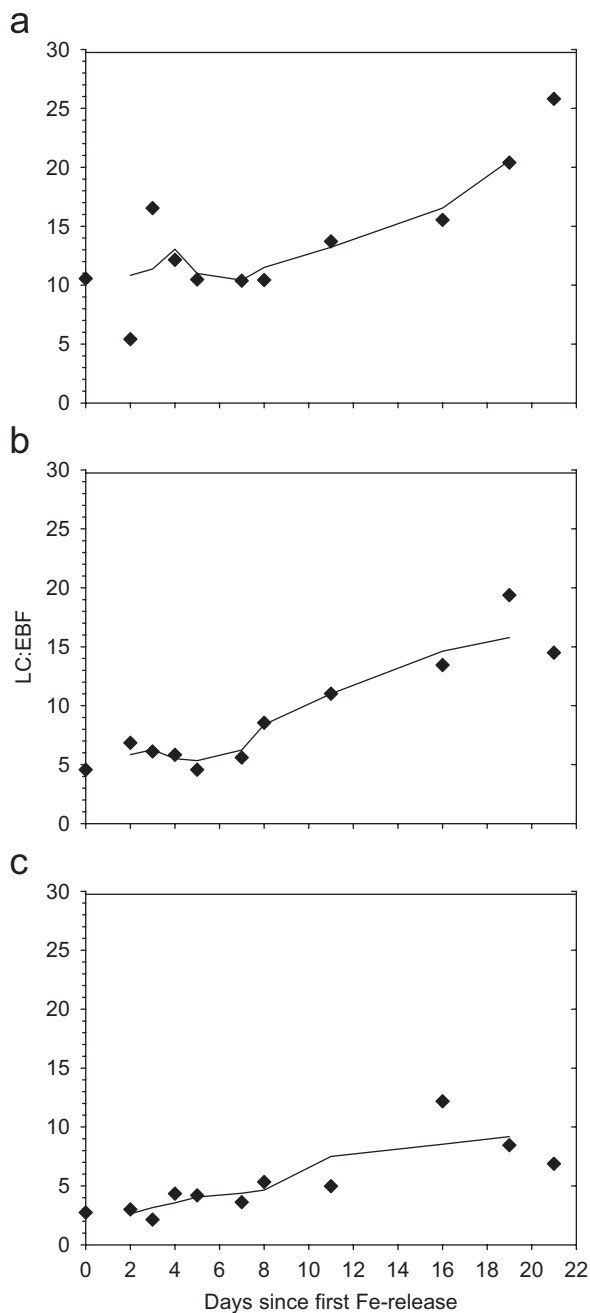


Fig. 6. Species-specific ratios of full cells vs. empty and broken frustules (LC:EBF) in three *Pseudo-nitzschia* species inside the patch. (a) *P. heimii*, (b) *P. lineola* and (c) *P. turgidula*. Thin line as in Fig. 1.

tentative conclusions regarding the relationship between proximate and ultimate mechanisms in shaping the structures of phytoplankton assemblages.

4.1. Significant species

Only 10 species significantly increased net accumulation rates in response to alleviation of iron limitation. These significant species are phylogenetically far apart and differ widely in morphology and size. The following brief descriptions (Hasle and Syvertsen, 1996) highlight the variety of shapes and sizes represented by five of the species from which their implications for food web relationships and for biogeochemical cycles can be derived.

Cells of *Pseudo-nitzschia lineola* are needle-shaped (56–112 by 1.8–2.7 μm), smooth, weakly silicified and attached at their tips in straight chains. *Fragilariopsis kerguelensis* cells are boat-shaped (10–76 by 5–11 μm), heavily silicified and attached along their entire lengths (the valve faces) in ribbon-like chains of 2–>100 cells. *Thalassionema nitzschioides* cells are rod-shaped (10–110 by 2–4 μm), heavily silicified (particularly the Antarctic strain encountered here) (Zielinski and Gersonde, 1997), and attached at the tips in rectangular, zigzagged chains. *Chaetoceros debilis* cells are cushion-shaped (8–40 μm), weakly silicified and with thin, long, curved setae protruding outward from the four corners and attached in characteristically spiraling chains. *Corethron pennatum* cells are large, cylindrical (20–200 by 5–40 μm), thin-walled, solitary and crowned with long, stout, barbed spines at both ends reminiscent of the spokes of a half-opened umbrella. Given the broad range of morphologies it is easy to envision that the subsequent fate of bloom biomass will depend on the species populations dominating it. Interestingly, the distribution ranges of these species, which reflect adaptation to specific proximate conditions or to water masses, also vary widely (Hasle and Syvertsen, 1996). *F. kerguelensis* is the most restricted and abounds only in the ice-free, open ACC, whereas *C. pennatum* is abundant throughout the Southern Ocean. The other species are cosmopolitan and occur from near-shore to open ocean environments. However, the heavily silicified form of *T. nitzschioides* predominates in the Southern Ocean (Zielinski and Gersonde, 1997).

In contrast, many of the background species present during EisenEx are endemic to the ACC. Some such as *Thalassiothrix antarctica*, *Corethron inerme*, and a few large *Chaetoceros* and *Thalassiosira* species have been reported as significant species in ACC blooms (Smetacek et al., 2004), but the majority seem to thrive at levels that were

background to the bloom—a few thousand cells or less per liter. The muted seasonal population dynamics of these background microplanktonic diatoms is reflected in abundances varying within an order of magnitude whereas the boom-and-bust outburst of bloom-forming species span 2 or more (Smetacek et al., 2004). Since congeneric species of the significant were also amongst the background species, high or low net accumulation rates were not related to morphology or phylogeny.

4.2. Growth conditions (bottom-up factors)

The phytoplankton community at the start of the experiment was dominated by the ubiquitous pico- and nanophytoplankton characteristic of HNLC oceanic waters that accounted for about 90% of the initial chlorophyll standing stock of $0.5 \mu\text{g l}^{-1}$ (Gervais et al., 2002). The diatom assemblage prior to fertilization was species-rich (> 50 species) but low in number (maximum $30 \times 10^3 \text{ cells l}^{-1}$). The experiment coincided with the seasonal spring increase, reflected in the concomitant, albeit modest diatom increase in surrounding waters indicating that phytoplankton growth rates were already iron limited at the start of the growth season. Bouts of stormy weather which homogenized the active mixed layer to the depth of the seasonal mixed layer at 80 m in the second half of the experiment did not constrain build-up of phytoplankton biomass (Fig. 1b) reflected in depth profiles of total diatom abundance (Fig. 2a) inside the patch. This regular deep mixing under generally cloudy skies will have exposed the entire phytoplankton assemblage to average light intensities of $66 \mu\text{mol m}^{-2} \text{ s}^{-1}$; (Gervais et al., 2002) so it is likely that the population as a whole will have been shade-adapted.

Macronutrients were still well above limiting concentrations at the end of the experiment (Bozec et al., 2005) and their availability should therefore not have impacted the observed successional patterns. Iron fertilization led to a significant increase in photosynthetic efficiency (F_v/F_m ratios) that continued to increase during the experiment (Gervais et al., 2002) and algal growth rates were estimated to be close to the maximum in the center of the patch (Bakker et al., 2005). Fe(II) levels were high throughout the experiment (Croot et al., 2005) so increasing F_v/F_m ratios can be explained by the growing proportion of cells with high ratios rather than by a general elevation of efficiency of the entire

assemblage. This implies that many cells did not respond to iron fertilization by elevating photosynthetic efficiency, a prerequisite to increasing growth rates. It is possible that other limiting factors such as cobalt and zinc constrained growth rates of some species. However, Scharek et al. (1997) observed no effect of supplementing these trace elements with iron additions to natural ACC phytoplankton populations.

The continuous exponential increase of *P. lineola* and *Chaetoceros debilis* populations through the experiment can be taken as proof of the temporal coherence of the in-patch stations. However, their gross accumulation rates (net growth rates) will have been about two-fold higher taking the effects of dilution into account. Sinking out of live phytoplankton cells and chains during the experiment can be considered negligible as evidenced by vertical profiles of FC (Fig. 2a) that showed a sharp decrease below the mixed layer. In contrast, profiles of EBF (Figs. 2b and c) did not exhibit a comparable marked decline below 80 m indicating that these particles were indeed sinking out of the surface layer. However, they contributed only a minor fraction of the total suspended particles, so sinking losses in general were low. Particle loss rates assessed by thorium measurements were also low and did not differ between patch and outside water (Rutgers van der Loeff and Vöge, 2001). This is in good agreement with results from the Southern Ocean Iron RElease Experiment (SOIREE) where reduced diatom sinking rates in response to iron addition were found and calculated sinking losses of iron-saturated, unaggregated cells were estimated to be $\sim 1\% \text{ d}^{-1}$ (Waite and Nodder, 2001). The longevity of the SOIREE bloom in surface waters of the ACC as indicated by satellite images (Abraham et al., 2000) further emphasises that sinking was a minor loss term.

4.3. The mortality environment (top-down factors)

Selective mortality will certainly be of importance in determining species composition of any bloom. Phytoplankton mortality is difficult to quantify in the field because it can result from a number of independent causes ranging from natural cell death (Bidle and Falkowski, 2004; Franklin et al., 2006), infection by viruses (Fuhrman, 1999; Tarutani et al., 2001; Brussaard, 2004; Bettarel et al., 2005), cell lysis (Berman and Wynne, 2005), pathogenic bacteria (Mayali and Azam, 2004) and parasitoids

(Park et al., 2004) to grazing by proto- and metazooplankton (Tillmann, 2004; Smetacek et al., 2004). In the case of diatoms, all causes of mortality will result in accumulation of EF with the notable exception of crustacean grazing that can produce both EF and BF. Heterotrophic dinoflagellates are specialized grazers of diatoms (Jacobson and Anderson, 1986; Siano and Montresor, 2005) and therefore a potential source of EF. Especially pallium-feeding dinoflagellates of the genus *Proto-peridinium* are reported to preferentially feed on diatoms (Menden-Deuer et al., 2005; Buskey, 1997). However, *Proto-peridinium* spp. accounted for only 4% of total dinoflagellate and less than 1% of total microzooplankton biomass (Henjes et al., 2007) and were likely only a minor cause of mortality. Another source of EF, not necessarily attributable to grazing, were EF within diatom chains. These were particularly frequent in *Fragilariopsis kerguelensis*, where 25–94% of EF were either present in empty chains or chains containing some full cells. However, studies on the feeding behavior of a small heterotrophic dinoflagellate (*Proto-peridinium vorax*) showed that it fed also on single or a few cells within a diatom colony (Siano and Montresor, 2005). Gametogenesis during sexual phases can also lead to EF (Crawford, 1995) but gametangia were rare during EisenEx.

The major cause of removal of EF from the mixed layer will be metazooplankton grazing because packaging of empty frustules in fecal pellets increases sinking rates by at least an order of magnitude. Furthermore, to our knowledge only crustacean zooplankton—copepods with mandibles and euphausiids with gizzards—have the ability to break diatom frustules. However, a considerable fraction of copepod and euphausiid feces is retained within the mixed layer (Gonzalez, 1992; Dagg et al., 2003), so their disintegration will result in release of EF and BF and occasionally full cells contained in them. Especially the cyclopoid copepod genus *Oithona*, numerically dominant during EisenEx (Henjes et al., 2007), feeds on fecal material. Gonzalez and Smetacek (1994) concluded that all fecal pellets are likely utilized in the upper 100 m if the ratio of cyclopoid to calanoid copepods exceeds four. The ratio of *Oithona* to all other copepods varied between 1.1 and 10.3 during EisenEx indicating that at high *Oithona* standing stocks a considerable fraction of fecal pellet production could have been retained within the mixed layer. So only the frustules crushed beyond recognition

and those removed via sinking feces will not be accounted for by this method. Selective loss of EBF of certain species via sedimentation of fecal pellets is unlikely since this would imply that fecal pellets with a certain species composition would preferentially sink out of the mixed layer as compared to fecal pellets with a different species composition.

The percentage of frustules crushed beyond recognition will vary between species and grazer. For instance, robust frustules such as those of *Fragilariopsis kerguelensis* are likely to withstand crushing by smaller copepods (Hamm et al., 2003) better than those of *Pseudo-nitzschia* which again will perform better than those of *Chaetoceros debilis*. So the EF:BF ratio provides a conservative estimate of the contribution of crustacean grazing to total mortality which will vary with the species-specific degree of robustness of the frustule. A considerable fraction of biogenic silica dissolves in the surface layer (Nelson et al., 1995), so selective dissolution likely also plays an important role in shaping the EBF assemblage. Because of the limitations listed above, enumeration of EBF underestimates diatom mortality and can only provide qualitative information on the mortality environment but nevertheless provides insight into grazer selectivity and trophic interactions.

The dynamics of microzooplankton and small copepod populations during EisenEx have been presented by Henjes et al. (2007). The biomass of larger protozooplankton capable of feeding on diatoms (ciliates, athecate and thecate dinoflagellates) increased from 653 to 1149 mg C m⁻² till the middle of the experiment and declined thereafter presumably because of selectively heavy grazing pressure on them by copepods (Henjes et al., 2007; Schultes et al., 2006). The numbers of copepodites and small copepods combined increased significantly in the course of the experiment inside the patch from 20 to a maximum of 36 individuals l⁻¹ (Henjes et al., 2007). These concentrations are in the upper range of values reported, even from coastal waters (Turner, 2004), so the grazing pressure of copepodites and small copepod species will have been heavy by any standard. Acoustic mapping of the fertilized patch suggests a three and five-fold increase in zooplankton abundance in the 120 and 38 kHz frequencies, respectively, presumably by congregation of adults of the larger (>2 mm) copepods (*Calanus similimus*, *Calanus propinquus*, *Calanoides acutus*, *Metridia* spp., *Pleuromamma robusta* and *Rhincalanus gigas*) from outside water

by regulation of vertical migration during the experiment (Krägersky et al., in prep.). This behavior has been identified as an important mechanism for vertically migrating zooplankton to congregate in patches of high food concentrations (Donaghay et al., 1991; Boyd et al., 1999). The resultant increase in copepod grazing pressure in the course of the experiment is indicated by the steady increase of fecal pellet carbon in the upper 80 m of the fertilized patch—from 81 to 203 mg C m⁻² (Henjes et al., 2007)—but no increase in surrounding waters.

Had the copepods grazed indiscriminately on suspended particles, as the much larger krill apparently do, mortality of the most abundant species—large diatoms—would have been highest. But the steady increase in the FC:EBF ratio of the dominant diatoms indicated that none of the dominant grazers fed selectively on them but preferred other species whose populations declined as a result. Thus, heterotrophic protists decreased during the second half of the experiment, following initial increases in population sizes. The impact of selective grazing pressure on this group was demonstrated by tintinnid ciliates whose empty loricae (including deformed and shattered loricae), like diatom frustules, enable assessment of mortality. Empty loricae increased from 348 to 670 l⁻¹ in the course of the experiment concomitant with a decline in the ratio full cells to empty loricae from 0.3 to 0.1 (Henjes et al., 2007). We cannot exclude that disruption of the ciliate cell by the fixation method or by viral lysis caused empty loricae but frequent observations of deformed loricae both freely suspended as well as in copepod fecal pellets suggests crustacean grazing as an important mortality term. In contrast to the significant diatoms, whose FC:EBF ratios ranged between 2 and 14, tintinnid mortality rates exceeded growth rates, so it is reasonable to expect the same of other, untraceable groups and species that did not contribute to the increase in total biomass.

Selective grazing on the faster-growing protozoa by the comparatively large stocks of copepods will have released grazing pressure on large diatoms (Henjes et al., 2007). We assume that protozoan grazing was the main cause of diatom mortality other than copepods because net accumulation rates of EF were higher in the upper layer, where the bulk of the protozooplankton population was located. Even within the large diatoms, grazing was selective as evidenced by the wide range in species-specific FC:EBF ratios.

4.4. Growth versus mortality rates

The ratio of growth to mortality rates determined species dominance during the iron-induced bloom assuming that none of the species populations present was severely resource limited, sinking losses were only of minor importance and dilution affected all species in the same manner. Although the EBF derived mortality rates underestimate in situ mortality rates because of the limitations listed above, the ratio of FC:EBF gives a reasonable indication which of the two processes dominated. In all of the significant species FC:EBF by far exceeded one, whereas in the background species FC:EBF was close to or below one indicating that in the former species growth dominated the observed net accumulation rates. Only two species from disparate genera, *Pseudo-nitzschia lineola* and *C. debilis*, maintained exceptionally high net accumulation rates throughout the experiment. The vast majority of species populations stayed well below the upper limit set by the two fast-growing species and remained either constant, grew for only a short period, or continuously albeit at a linear rate. These populations were either controlled by higher selective mortality or did not adjust their gross growth rates in response to iron addition. The differences in the dynamics of phytoplankton species populations recorded during EisenEx, differentiated into six categories, reflect the range of growth strategies represented in the phytoplankton.

Both *P. lineola* and *C. debilis* exhibited remarkably constant net accumulation rates throughout the experiment of about 0.2 d⁻¹ that were double those of most other dominant species. Furthermore, taking patch dilution (~0.2 d⁻¹) and EBF derived mortality rates (see Section 3) into account we estimate a gross growth rate for *P. lineola* of ~0.45 d⁻¹ and a similar growth rate can be assumed for *C. debilis*. This assumption is conservative since the EBF-derived mortality will underestimate the true mortality rates. The difference in contribution to bloom biomass between these two fast-growing species was a function of initial seeding stock. *P. lineola* started with an order of magnitude higher inoculum as compared to *C. debilis* and therefore dominated bloom biomass inside the patch after 3 weeks. The exponential increases of both fast-growing species demonstrate the potential of diatom growth physiology despite the suboptimal light levels to which most of the population in the 80 m mixed layer was subjected on a daily basis.

Adaptation to deep mixed layers cannot be invoked since both species have widespread distributions (Hasle, 1965; Fryxell and Medlin, 1981) indicating a corresponding broad range of adaptation to environmental variables. Further, *C. debilis* dominated to 90% the iron fertilization experiment SEEDS carried out in the HNLC NE Pacific at 10 °C in a 10 m mixed layer (Tsuda et al., 2005). Clearly, the high growth rates of *C. debilis* are not restricted to a narrow range of environmental conditions. Unfortunately mortality rates during EisenEx could not be estimated from assessment of EBF because of their fragility.

Outside the patch, *P. lineola* was one of six species that increased abundance during the 3 weeks, indicating that even under iron-limiting conditions it had higher growth, lower mortality rates, or both, than other species. So, it cannot be classed as a species requiring high iron concentrations for it to grow. Some species of the genus *Pseudo-nitzschia* produce the neurotoxin domoic acid which is known to bind iron and might be of competitive advantage under iron limitation (Maldonado et al., 2002). However, domoic acid was not found in bulk plankton samples from EisenEx (Lucas, M., pers. comm.). The consistently higher FC:EBF ratio in *P. lineola* as compared to most other dominant species, suggest that its dominant status was achieved primarily by exceptionally high, sustained gross growth rates. This conclusion is supported by comparison with other *Pseudo-nitzschia* species.

The fact that morphologically similar species of the genus *Pseudo-nitzschia* exhibited consistently differing patterns in FC:EBF and EF:BF ratios suggests that their grazers, for unknown reasons, were discriminating between species and not cell shape because they are morphologically very similar. These differences in congeneric species cannot be attributed to methodological bias induced by differences in fragility and robustness of the frustules as can be expected in the case of species as different as *F. kerguelensis* and *C. pennatum*. *Pseudo-nitzschia heimii* had lower mortality rates than the two sibling species and grazing pressure on *P. turgidula* increased in the latter half of the experiment. Decreasing EF:BF ratios in *P. turgidula* point to crustacean grazing as the main mortality cause in this species. The fact that *P. heimii* suffered lower mortality than *P. lineola*, as indicated in the consistently higher FC:EBF ratios, but did not contribute significantly to bloom biomass, further highlights the exceptional growth performance of *P. lineola*.

The large, solitary diatoms *C. pennatum* and *Haslea trompii* contributed significantly to diatom biomass despite relatively low abundances. Gross growth rates (estimated from net accumulation rates after correcting for dilution and EBF-derived mortality) of $\sim 0.31 \text{ d}^{-1}$ for *C. pennatum* were lower than those of the two fast-growing species. The comparative paucity of empty frustules of *C. pennatum*, despite their conspicuousness, indicate that large size and long stout, barbed spines provided effective protection against protozoan grazers, especially ciliates. However, mortality due to copepod grazing on this species was higher than in all other large diatoms reflected in the low FC:EBF ratios of 1.8–5.2. Indeed, *Calanus simillimus* preferentially fed on *C. pennatum* in bottle experiments (Schultes et al., 2006). Nevertheless, the linear increase in population size in the course of the experiment indicates either decreasing growth or increasing mortality rates over time.

High net accumulation rates of *Fragilariopsis kerguelensis* and *Thalassionema nitzschioides* lasted for 9 and 14 days, respectively in the middle of the experiment and resulted in a four-fold increase in abundance of both species. In this period, net accumulation rates of 0.15 d^{-1} for *F. kerguelensis* were the third highest recorded during EisenEx indicating that thick frustules do not imply low division rates. Taking net accumulation rates, dilution and EBF-derived mortality rates into account, gross growth rates for *F. kerguelensis* and *T. nitzschioides* during the period of spurt growth were ~ 0.38 and $\sim 0.30 \text{ d}^{-1}$, respectively. So why did population sizes not change during the first and last third of the experiment? Differentiating protist from copepod grazing rates based on EF:BF ratios are difficult to assess for these species because their robust, heavily silicified frustules can withstand copepod handling (Hamm et al., 2003), as indicated by numerous intact empty and even full cells in their fecal pellets (Jansen, 2006). The increasing FC:EBF ratio until the end of the experiment, accompanied by only slight increases in EF and BF, indicates that relative increases in mortality rates could not be the primary cause for the periods of static population size of *F. kerguelensis*. Moreover, the rapid increases in population sizes occurred in the middle of the experiment when protistan grazing pressure peaked and that of copepods was steadily increasing. Hence, it appears that periods of relative inactivity were interspersed with phases of rapid growth. Sinking losses in *F. kerguelensis* can be regarded as

negligible as also reported for the SOIREE bloom (Waite and Nodder, 2001).

In the case of *T. nitzschioides*, the low net accumulation rates during the first week of the experiment are likely to have been due to grazer control reflected in the low LC:EBF ratios of 1.5 and 1.2 during day 2 and 3, respectively. The rod-shaped cells of *T. nitzschioides* are rather small and therefore accessible also to small protozoan grazers that are able to ingest cells and digest the contents without breaking the robust frustule. The initial increase in protozoan grazers suggests that they might have controlled the *T. nitzschioides* population during the initial phase of the experiment. In vitro grazing rates of larger copepods on *T. nitzschioides* were lower than on other diatoms (Schultes et al., 2006).

Phototrophic dinoflagellates (mainly *Prorocentrum* spp.) and various small to medium sized diatoms such as *Cylindrotheca closterium* declined during the second half of the experiment. Frequent observations of recognizable remains of *Prorocentrum* spp. in copepod fecal pellets suggest that crustacean grazing was largely responsible for their decline. The different abundance patterns observed for solitary cells and colonial *Phaeocystis antarctica*—a decline in the former and an increase in the latter—illustrate how grazing can structure species populations. The decline in solitary cells within the patch is most likely due to grazer-induced mortality. Especially ciliate grazers are known to rapidly ingest solitary cells (Verity, 2000) and the increase in aloricate ciliate abundance inside the patch (Henjes et al., 2007) suggests that single cells were likely controlled by ciliate grazers. The tough skin of *P. antarctica* colonies on the other hand provides protection against various pathogens (Brussaard et al., 1999) and predators (Verity, 2000) and can be viewed as a flagellate's way of surviving in the mortality environment (Hamm et al., 1999).

Species of category 5 that showed no response to fertilization were generally of large size and present in similarly low abundances both inside and outside the patch. The relatively stable populations of these “background” species, including *Thalassiosira lentiginosa*, which is prominent in underlying sediments, were either regulated by heavy grazing pressure or maintained low gross growth rates despite sufficient resources. The FC:EBF ratios in the background species varied between 1 and 1.6 in the case of *Asteromphalus hookeri* and *T. lentiginosa*, respectively and 7.2 in the case of *Thalassiosira oestrupii*

indicating that grazing pressure on the two former species was higher as compared to the latter species. However, some of the species in this category have made significant contributions to natural blooms (mentioned above), so low growth rates during EisenEx cannot be attributed to inherently inferior resource-uptake efficiency in comparison with species of previous categories. Unfavorable growth conditions cannot be invoked either as most of these species are endemic to the ACC, implying that they are specifically adapted to the environmental conditions. Furthermore, several species of this category were congeneric to the fast-growing ones so it is unlikely that the physiological potential for growth would have been very different within the respective genera. We suggest that low growth rates during the experiment are more likely due to internal regulation than to species-specific, innate physiological constraints.

Internal regulation implies the presence of life-cycle controlling genes. Long-term datasets from iron- and nutrient-replete coastal regions indicate that the timing of species appearance in the course of annual succession cycles is more dependent on the calendar than on proximate environmental conditions (Ribera D'Alcalá et al., 2004). The latter determine growth performance but initiation of higher growth rates in the population appears to be controlled internally, whether triggered by external cues such as day length or temperature or by internal clocks as reported from an aquatic macrophyte (Huebert and Gorham, 1983) and many land plants (John and Nadgauda, 1999). So it is possible that one or another of these species would have significantly increased growth rates had the experiment been prolonged. The point is that replete resources are an insufficient condition to trigger maximum growth rates in phytoplankton populations.

The significant, steady decline in the coccolithophorid *Emiliania huxleyi* inside and outside the patch at the time of vernal increase suggests that its net accumulation rate was constrained by factors other than resources. Coccolithophore abundances are known to decrease drastically from the Sub-tropical Convergence to the Polar Front (Findlay and Giraudeau, 2000; Nishida, 1986) presumably because of temperature limitation of growth rates. It is hence likely that the population present in the region had been recently advected southward from sub-Antarctic waters. Hydrographical mapping of the eddy prior to fertilization provides support for

this scenario (Cisewski et al., 2005). Grazing mortality of *E. huxleyi* by microzooplankton exceeded population growth rates as shown by serial dilution experiments (data not shown) conducted during EisenEx. The low FC:EBF ratio of 0.3 for another traceable species, *Fragilariopsis rhombica* in this category, showed a similar decline in abundance supporting the role of grazer control.

4.5. Implications for evolutionary ecology and biogeochemistry

The exceptionally high growth rates of two diatom species from disparate genera demonstrate the potential efficiency of resource utilization that can be achieved by diatom ecophysiology. The fact that only two species attained and maintained these rates implies that all the other diatom species, whether endemic or cosmopolitan, were growing below or well below the rates set by ecophysiological constraints of the lineage. Since closely related but slower growing species from both genera were also present, one cannot conclude that evolution of the respective lineages is driven solely by resource competition, as this would imply predominance of bloom-forming species. This phylogenetic diversity within ecological groupings has interesting implications for our understanding of the evolutionary ecology of marine phytoplankton. Indeed, the fact that the vast majority of phytoplankton species do not contribute significantly to blooms indicates that the density of individuals required for a species population to survive and evolve, i.e. to maintain fitness in the marine plankton, is two orders of magnitude below that achieved by bloom-forming species. Clearly, the latter <10% of the species pool have not out-competed the >90% background species over evolutionary time scales, implying that the life cycle with a bloom-forming stage is one of many viable life cycle strategies in the phytoplankton (Smetacek et al., 2004). These results cast doubt on the selection of fast growth rates as the driving force in plankton evolution implying that resource-based competitive exclusion is probably not the rule in pelagic ecosystems.

The role of mechanical defense in shaping the evolutionary ecology of phytoplankton has been discussed by Smetacek et al. (2004). Indeed, it is no coincidence that the vast majority of bloom-forming phytoplankton have tough cell walls of various materials: silica, carbonate or organic scales or plates. Clearly, some degree of mechanical protec-

tion is a prerequisite to bloom formation and diatoms seem to have evolved the cheapest armor, with immense consequences for the organic carbon pump and hence global biogeochemical cycles (Smetacek, 1999).

5. Conclusions

The need to address the species level in biogeochemical studies is illustrated by the following thought experiment: If all viable diatom cells at the start of EisenEx had accumulated at rates of the two fast-growing species (*Pseudo-nitzschia lineola* and *C. debilis*), the diatom biomass we measured on day 21 would have been reached by day 8 and on day 21 it would have attained 45 g C m^{-2} instead of the 3.5 g C m^{-2} we recorded. Conversely, had *P. lineola* been absent, diatom biomass on day 21 would have been less by 25%. Our observations do not suggest that its place would have been taken by another species. Clearly, one cannot assign a universal, in situ growth rate to “diatoms” in general, based only on temperature and resource availability. The need to further differentiate “functional groups” into dominant species should be obvious.

One cannot invoke species-specific adaptation to a finely tuned set of environmental variables such as light climate or nutrient ratios (Sommer, 1999) as both the fast-growing species are cosmopolitan and have dominated blooms under very different conditions. Besides, storm-induced variation in light climate had no effect on the accumulation rates of the exponential species. Most species apparently regulated their growth rates at levels below those of the bloom-formers or were kept in check by the grazer community. Clearly, sufficient light and nutrient conditions are a prerequisite for bloom formation but its rate of biomass build-up will depend on the number of activated, bloom-forming species present, their initial population size (seeding cells), selective grazing pressure and other sources of mortality. It follows that the development of phytoplankton blooms will as much depend on ultimate mechanisms (e.g. growth regulation and internal clocks geared to the season) as on proximate environmental conditions (e.g. resource availability and grazing pressure). Hence, the relationship between the two will determine the magnitude, timing, composition and fate of phytoplankton blooms. This is illustrated by a comparison with other Southern Ocean iron fertilization

experiments that differed not only in length, season and depth of the mixed layer but also in the composition and seeding concentration of the respective phytoplankton assemblages (Gall et al., 2001; Coale et al., 2004; Hoffmann et al., 2006). The response categories as described above will be there in each case but the species constituting them will be different.

We do not know the fate of the EisenEx bloom but speculate that the *Pseudo-nitzschia lineola* population is unlikely to have been grazed down but will have crashed at some point, whether triggered by external (e.g. nutrient depletion, pathogens) or internal (e.g. programmed cell death, mass sexual phase) factors. Our view is supported by the fact that both *Pseudo-nitzschia* and *Chaetoceros* blooms are known to form aggregates and sink out en masse (Alldredge and Gotschalk, 1989; Stachowitsch et al., 1990; Buck and Chavez, 1994; Alldredge et al., 1995; Parsons and Dortch, 2002). We therefore assume that a significant fraction of the biomass of these bloom-forming species will have settled out of the surface mixed layer and transported carbon to the deep sea.

Our observations highlight the role of species-specific internal regulation of growth rates in determining the dynamics of phytoplankton blooms. Rapid advances in genomics lead one to anticipate identification of the genetic correlates of phytoplankton behavior patterns observed in the field. Elucidating the other side of the coin—the mechanisms of evolutionary selection of species-specific morphological and behavioral traits—will require development of a complex, coherent framework of phytoplankton ecology on a par with that developed for land plants. On land, growth is only a part of the ecological framework: chemical (toxins) and mechanical (thorns, bark) defense, pollination (sex), seed dispersal are equally important factors shaping species-specific patterns of occurrence and their respective impacts on ecosystem structure and functioning.

Why should phytoplankton evolutionary strategies accompanied by the inevitable trade-offs be less complex given that the various phylogenetic groups comprising marine phytoplankton share evolutionary histories equivalent to those of land plants? The rate of speciation in some diatom genera rivals or even exceeds that of rapidly evolving land plant lineages such as grasses (Kooistra and Medlin, 1996). So one should anticipate equivalent differences in adaptive strategies to ecological conditions

amongst diatom species as have been established for angiosperms (grasses, herbs, shrubs and trees) by terrestrial ecologists. In situ fertilization experiments represent an ideal method to advance our understanding of oceanic phytoplankton ecology.

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